

PUNE, India

WMO: 430630

Lat: 18.533N

Lon: 73.850E

Elev: 1831

StdP: 13.75

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	49.9	52.1	34.7	31.5	88.4	38.1	36.0	86.1	7.7	79.0	6.9	78.6	0.0	90	0.260

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	22.6	101.3	67.8	99.4	67.7	97.5	67.9	76.5	86.0	75.8	84.7	75.2	83.7	3.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	137.6	79.2	73.7	134.3	78.5	73.1	131.5	78.1	41.6	86.4	40.8	85.3	40.1	83.8	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	8.8	7.3	45.7	104.9	2.0	1.5	44.3	106.0	43.1	106.8	42.0	107.7	40.5	108.8		
			43.8	78.9	2.0	2.0	42.4	80.3	41.2	81.5	40.1	82.7	38.6	84.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.9	69.0	73.1	79.0	84.6	85.9	81.1	77.1	76.2	77.2	77.3	73.1	69.6
	DBStd	5.88	3.27	3.49	3.68	3.08	2.59	3.51	2.13	1.68	2.02	2.70	3.35	3.40
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	11	7	1	0	0	0	0	0	0	0	0	0	3
	CDD50	9834	589	647	900	1039	1112	932	839	812	816	847	694	607
	CDD65	4370	131	228	435	589	647	482	374	347	366	382	244	145
	CDH74	43811	1637	2869	5828	8107	8219	4373	1950	1473	2138	3215	2352	1650
	CDH80	19813	461	1338	3404	4945	4594	1604	293	162	525	1257	804	426
Wind	WSAvg	2.2	0.9	1.3	1.5	2.2	4.1	4.2	3.9	3.4	2.2	1.0	0.9	0.8
Precipitation	PrecAvg	28.7	0.0	0.0	0.2	0.4	1.3	5.9	7.2	5.4	5.3	3.0	1.0	0.2
	PrecMax	52.9	0.7	1.2	2.3	2.5	5.9	19.7	16.2	14.9	17.0	10.6	7.0	2.8
	PrecMin	15.1	0.0	0.0	0.0	0.0	0.0	0.5	1.3	0.5	0.9	0.0	0.0	0.0
	PrecStd	8.4	0.1	0.2	0.4	0.6	1.4	3.9	3.6	3.2	3.5	2.7	1.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.0	95.0	100.6	103.9	103.7	97.9	87.9	86.1	89.5	92.1	89.6	88.0
		MCWB	64.6	64.0	65.4	66.7	69.3	74.0	74.2	74.7	73.4	69.9	68.2	66.4
	2%	DB	86.6	92.5	98.5	101.7	101.2	94.2	85.0	83.5	87.1	90.0	87.9	85.8
		MCWB	64.2	63.9	65.4	67.0	69.5	74.1	74.0	73.9	74.2	69.9	67.8	65.7
	5%	DB	84.8	90.4	96.5	99.9	99.0	91.2	83.2	81.8	85.1	88.4	86.3	84.2
		MCWB	63.8	63.8	65.0	66.9	69.7	74.2	73.7	73.3	73.7	70.0	67.6	65.3
	10%	DB	82.7	88.1	94.5	98.1	96.7	88.4	81.4	80.3	83.2	86.7	84.7	82.5
		MCWB	63.6	63.8	64.4	66.9	70.7	74.3	73.5	73.0	73.3	70.1	67.4	64.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.2	69.5	72.8	74.7	77.1	77.6	76.2	76.3	77.1	76.5	74.5	72.0
		MCDB	81.9	85.5	84.3	88.7	90.8	87.8	82.4	82.6	85.1	84.2	82.6	80.8
	2%	WB	67.3	67.5	70.4	73.1	75.9	76.5	75.1	74.9	76.0	75.5	73.2	70.0
		MCDB	81.3	85.1	85.2	88.0	89.5	86.5	81.2	80.1	83.2	83.1	81.6	80.0
	5%	WB	65.9	66.4	68.6	71.8	74.9	75.7	74.5	74.2	75.1	74.6	72.1	68.5
		MCDB	80.5	84.4	86.1	87.8	88.6	85.1	80.2	79.3	81.7	81.9	79.9	79.0
	10%	WB	64.6	65.2	67.3	70.6	74.1	75.0	73.9	73.6	74.3	73.8	71.0	66.8
		MCDB	79.4	83.2	86.5	87.3	87.4	83.8	79.2	78.6	80.2	80.5	78.6	78.0
Mean Daily Temperature Range	5% DB	MDBR	28.9	30.9	31.5	28.5	22.6	13.2	8.1	8.5	12.7	19.8	24.9	27.8
		MCDBR	30.6	33.1	32.9	30.2	26.5	18.9	11.4	11.3	16.3	23.0	26.3	28.8
		MCWBR	10.8	9.9	9.0	8.8	7.5	4.4	3.8	4.1	5.4	6.9	9.1	10.7
	5% WB	MCDBR	27.0	28.7	29.2	27.0	21.6	14.6	8.8	9.2	13.3	16.6	18.5	23.6
		MCWBR	10.7	10.1	9.2	8.8	6.3	4.8	3.8	4.1	5.2	5.9	7.4	9.7
Clear-Sky Solar Irradiance	taub	0.424	0.431	0.499	0.550	0.543	0.578	0.576	0.544	0.522	0.554	0.494	0.441	
	taud	2.158	2.160	1.986	1.891	1.949	1.916	1.919	2.012	2.064	1.914	2.029	2.130	
	Ebn at Noon	269	275	260	247	246	235	236	246	250	236	246	259	
	Edh at Noon	45	47	58	65	60	62	62	56	53	60	51	45	
All-Sky Solar Radiation	RadAvg	1591	1862	2099	2236	2213	1493	1112	1146	1418	1617	1562	1510	
	RadStd	46	44	70	56	107	153	134	84	124	86	84	42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	+0.52	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.50	+0.94	N/A	+0.61	+0.43	+0.47	N/A	N/A	+288

Nomenclature: See separate page